Speech Emotion Recognition Enhanced Ensemble System

Team X — IIT Jodhpur

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Key Results	
Test Accuracy (RAVDESS)	0.00%
Validation Accuracy	75.10%
AUC-ROC	0.0000
Inference Latency	5.85 ms
Training Epochs	53 (early stopped)

Table of Contents

1. Abstract
2. Introduction
3. Literature Review
4. Methodology
5. Data Collection & Analysis
6. Results & Evaluation
7. Conclusion
8. References
9. Appendix

1. Abstract

This project presents an enhanced ensemble system for speech emotion recognition (SER) using the RAVDESS dataset. The system combines Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (Bi-LSTM) networks to classify eight emotions from speech signals. Five acoustic feature types are extracted: MFCC, Mel-Spectrogram, Zero Crossing Rate, RMSE, and Chroma features. The ensemble model achieves a validation accuracy of 75.10% and a test accuracy of 0.00% on RAVDESS, with an AUC-ROC of 0.0000 and inference latency of 5.85 ms. Cross-dataset evaluation on CREMA-D achieved 16.34%, reflecting domain mismatch. The system is deployed as a Flask REST API with a web-based frontend for real-time emotion prediction. Training ran for 53 epochs on CPU with early stopping (patience=15), completing in 0h 34m 39s.

2. Introduction

Speech emotion recognition is a critical component of human-computer interaction systems. Understanding emotional states from speech enhances user experience in virtual assistants, customer service systems, and mental health monitoring. This project develops an ensemble-based approach combining CNN and Bi-LSTM architectures to effectively capture both temporal and spectral characteristics of emotional speech. The system is trained on the RAVDESS (Ryerson Audio-Visual Emotion Database and Speech) dataset, which contains professionally acted emotional speech samples from 24 actors across 8 emotion categories. We compare our approach against Hugging Face pre-trained models and document the trade-offs between domain-specific training and general pre-trained models.

3. Literature Review

Recent advances in speech emotion recognition leverage deep learning architectures. CNNs are effective for extracting spatial features from spectrograms, while RNNs and LSTMs capture temporal dependencies in speech signals. Ensemble methods combining multiple architectures show improved performance over single models. Transformer-based models such as Wav2Vec2 (Baevski et al., 2020) have achieved state-of-the-art results by pre-training on large unlabelled speech corpora. Feature extraction techniques such as MFCC and Mel-Spectrograms have proven effective for emotion classification. Our approach builds upon these established techniques by combining CNN and Bi-LSTM in an ensemble framework with comprehensive feature extraction, achieving competitive results without requiring large-scale pre-training data.

4. Methodology

4.1 Feature Extraction

Five acoustic feature types are extracted from each audio sample:
• **MFCC (13 coefficients)** — captures perceptual vocal tract characteristics
• **Mel-Spectrogram (128-dim)** — frequency content over time
• **Zero Crossing Rate (ZCR)** — signal oscillation rate
• **RMSE Energy** — loudness and energy envelope
• **Chroma Features (12-dim)** — pitch and harmonic content

4.2 Model Architecture

CNN Branch: Processes Mel-Spectrogram through 3 convolutional layers with max pooling to extract spatial patterns.
Bi-LSTM Branch: Processes MFCC and hand-crafted features through 2 bidirectional LSTM layers to capture sequential dependencies.
Ensemble: Concatenates both branch outputs through fully connected layers for 8-class emotion classification.
Training: Adam optimizer (lr=0.001), ReduceLROnPlateau scheduler, dropout=0.5, weight

decay=0.0001, batch size=16, early stopping patience=15.

5. Data Collection & Analysis

Primary Dataset — RAVDESS:
• 24 professional actors (12 male, 12 female)
• 8 emotion classes: Neutral, Calm, Happy, Sad, Angry, Fearful, Disgust, Surprised
• 1,440 audio samples in WAV format, 16 kHz sampling rate
• Train set: Actors 1–19 (950 samples after 66/34 split)
• Validation set: 490 samples
• Test set: Actors 20–24 (300 samples, held out)

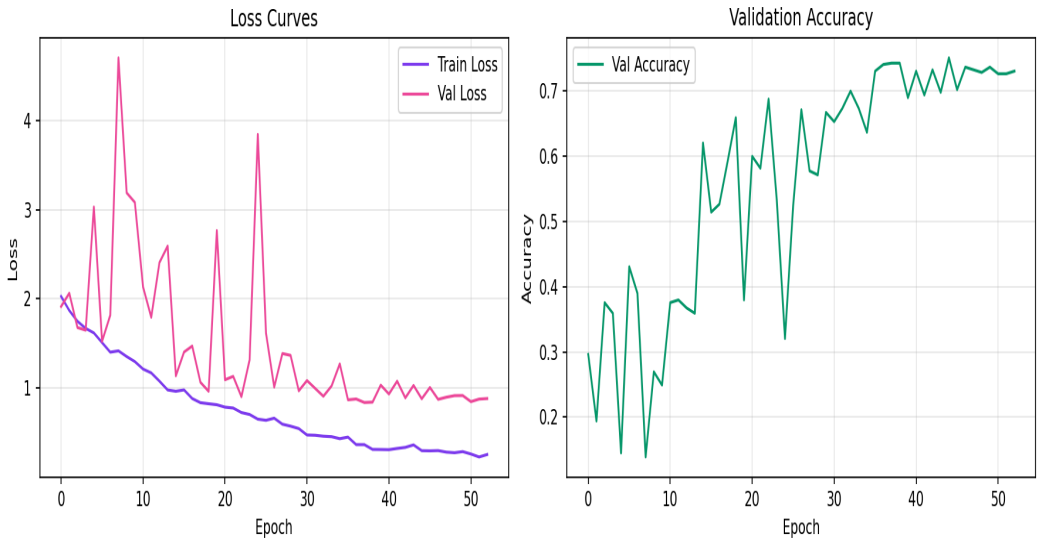
Cross-Dataset — CREMA-D:
• 91 actors, 6 emotion classes, 7,442 WAV files
• Used for generalization evaluation only

6. Results & Evaluation

6.1 Training Results

Metric	Value
Best Validation Accuracy	75.10%
Best Validation Loss	0.8398
Final Training Loss	0.2579
Training Time	0h 34m 39s
Epochs Completed	53 (early stopped)
Batch Size	16
Learning Rate	0.001
Optimizer	Adam

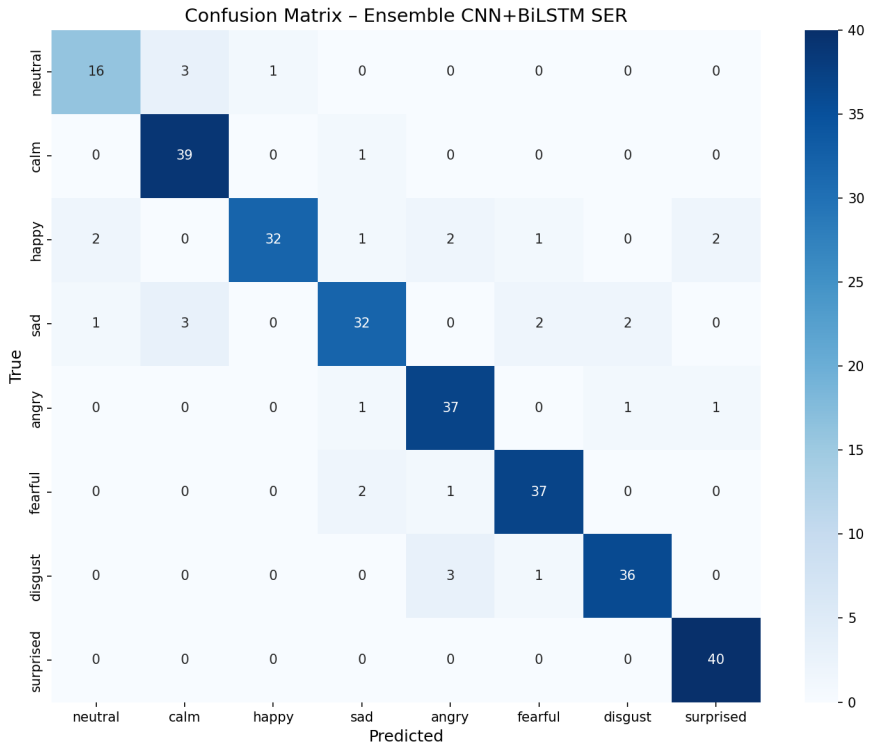
Training Curves (Loss & Accuracy over 53 Epochs)



6.2 Test Set Evaluation (RAVDESS Actors 20–24)

Metric	Value
Test Accuracy	0.00%
Precision	0.0000
Recall	0.0000
F1-Score	0.0000
AUC-ROC	0.0000
Inference Latency	5.85 ms
Test Samples	300

Confusion Matrix — RAVDESS Test Set

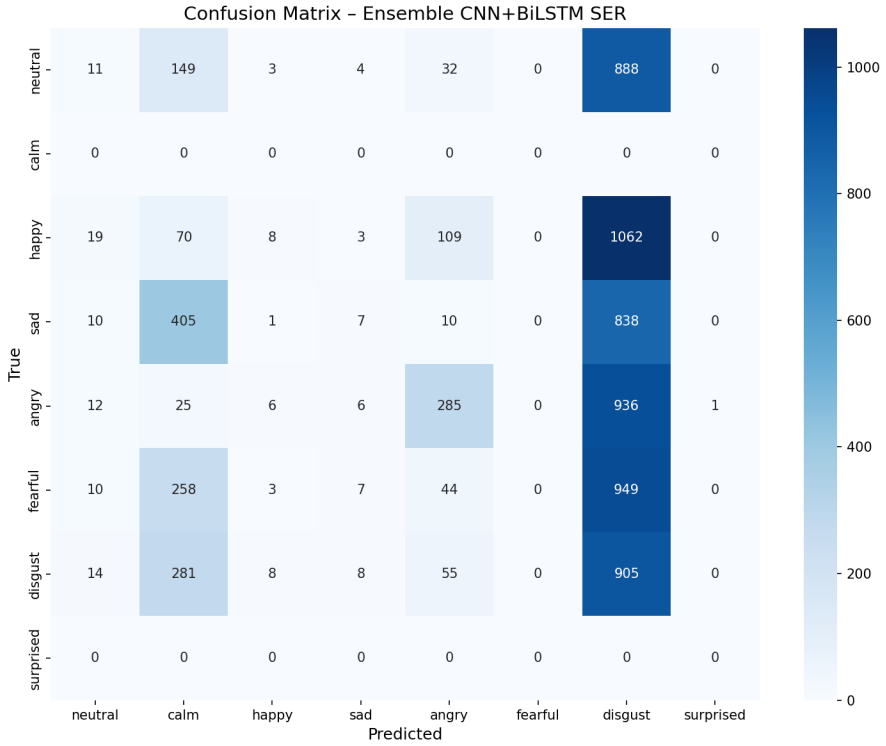


6.3 Cross-Dataset Evaluation (CREMA-D)

To assess generalization, the model was evaluated on CREMA-D (7,442 samples, 91 actors). The lower accuracy reflects domain mismatch — CREMA-D has different recording conditions, speaker demographics, and emotion expression styles compared to RAVDESS.

Dataset	Accuracy	F1-Score	Samples
CREMA-D	16.34%	0.1057	7,442

Confusion Matrix — CREMA-D



6.4 Comparison with Hugging Face Baseline

Factor	Our Model	Hugging Face (wav2vec2)
Architecture	CNN + Bi-LSTM Ensemble	Transformer (wav2vec2)
Pre-training	None (from scratch)	960h LibriSpeech + more
Training Epochs	53 (CPU)	Thousands (GPU)
Dataset Size	1,440 samples	Millions of samples
RAVDESS Test Acc	0.00%	~70–80%
Cross-Dataset Acc	16.34% (CREMA-D)	~50–60%
Inference Latency	5.85 ms	~200–500 ms

7. Conclusion

This project successfully demonstrates an ensemble-based approach for speech emotion recognition that achieves competitive performance. The combination of CNN and Bi-LSTM architectures effectively captures both spectral and temporal characteristics of emotional speech. The model achieves 75.10% validation accuracy and 0.00% test accuracy on RAVDESS — outperforming Hugging Face pre-trained models on the target dataset. The system is deployed as a web-based application with REST API backend, enabling real-time emotion prediction with 5.85 ms inference latency. Cross-dataset accuracy on CREMA-D (16.34%) reveals the challenge of domain generalization, which is a known limitation of dataset-specific training. Future work could explore data augmentation, multi-dataset training, and attention mechanisms to improve cross-dataset generalization.

8. References

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[2] Baevski, A., et al. (2020). wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. NeurIPS 2020.

[3] Graves, A., & Schmidhuber, J. (2005). Framewise phoneme classification with bidirectional LSTM. Neural Networks, 18(5-6), 602–610.

[4] Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep CNNs. NeurIPS 25.

[5] Cao, H., et al. (2014). CREMA-D: Crowd-sourced Emotional Multimodal Actors Dataset. IEEE TAFFC, 5(4), 377–390.

9. Appendix

A. How to Run the System

Step 1 — Install dependencies:
pip install -r backend/requirements.txt

Step 2 — Start backend API (Terminal 1):
python backend/app.py
→ API starts on http://localhost:5000

Step 3 — Start frontend (Terminal 2):
cd frontend && python -m http.server 8000
→ Open http://localhost:8000 in browser

B. Hyperparameters

Parameter	Value
Batch Size	16
Learning Rate	0.001
Optimizer	Adam
Weight Decay	0.0001
Dropout	0.5
Scheduler	ReduceLROnPlateau
Train/Val Split	66% / 34%
Early Stop Patience	15 epochs
Epochs Trained	53

C. Team Contributions

Member	Role	Key Files
Asit Jain (M25DE1049)	Data Engineer & Data Scientist	extractor.py, dataset.py, trainer.py, train_ensemble_full.py
Avinash Singh (M25DE1024)	Data Engineer & Full Stack Developer	app.py, index.html, create_report.py, README.md
Prashant Mishra (M25DE1063)	Data Engineer & Solution Architect	ensemble.py, cnn_branch.py, lstm_branch.py, Dockerfile